

Figure 4.5 starts off with \$1 at the beginning of January 1926. The dashed line represents a 100% bond position, which rises steadily. A 100% stock position is shown in the dotted line, and the stock wealth is relatively volatile. Stocks rise though the 1920s and reach a peak of \$2.93 at the end of August 1929. Then the Great Depression hits with a vengeance. Stocks markets crash and remain depressed into the early 1930s. Stocks hit a minimum of \$0.49 in May 1932. Stocks begin a slow climb upward from this point and end in December 1940 at \$1.81, which is below the cumulated bond position of \$2.08 at that time. The solid line in Figure 4.5, Panel A, shows the rebalanced 60%/40% position. It is much less volatile than the 100% stock position so, while it does not rise as much until 1929, it also does not lose as much during the early 1930s. The 60%/40% strategy ends at December 1940 at \$2.46.

Rebalancing is beneficial during the early twentieth century because it counter-cyclically cuts back on equities as they were peaking in 1929 and adds equities when they were at their lowest point in the early 1930s. Panel B of Figure 4.5 shows the rebalanced strategy, which goes back to 60%/40% at the end of each quarter, versus a buy-and-hold strategy, which starts off at 60%/40% at the beginning of the sample and then fluctuates only according to how bond and stock returns vary. The rebalanced strategy, by design, hovers around the 60% equity proportion. There are some deviations because the strategy is not continuously rebalanced, but overall *the rebalanced strategy is less risky* because it does not allow equities to rise or fall to dangerously high or low levels. In terms of utility, the rebalanced strategy attains the optimal balance of stocks and bonds for the investor's risk aversion. But, as an added benefit, rebalancing is counter-cyclical. In contrast, the equity holding in the buy-and-hold strategy was very high in early 1929 (when stock prices are high and expected returns low), right before stocks crash in October 1929. The buy-and-hold equity weight was very low in 1932, right before stock prices pick up (stock prices are low, and expected returns are high).

Rebalancing over 1990–2011

Figure 4.6 does a similar exercise for the 1990–2011 period. In Panel A, we start with \$1 invested at the beginning of January 1990. The bond position is shown in the dashed line. During 2008, bond prices suddenly spiked as there was a flight to quality when Lehman Brothers failed, but overall the series is relatively stable. The ending bond position at December 2011 is \$7.12. The equity position in the dotted line shows two large peaks and declines: the bull market of the late 1990s followed by the bursting of the Internet bubble in the early 2000s and the rise in equity prices during the early to mid-2000s followed by the financial crisis in 2007 and 2008. The ending equity position at December 2011 is \$6.10. Like Figure 4.5, the solid line shows returns of a rebalanced 60%/40% strategy where the rebalancing occurs at the end of every quarter. This dynamic strategy is less volatile, by holding fewer equities, than the 100% equity position, and ends up doing better at December 2011, at \$7.41, than either than full stock or bond strategy.